Product Requirements Document (PRD): CIRA – CIR Industry Readiness Accelerator

1. Executive Summary & Problem Statement

Executive Summary

The CIR Industry Readiness Accelerator (CIRA) is a comprehensive, university-wide platform engineered to transform the Career and Industry Readiness (CIR) lifecycle. CIRA provides a robust ecosystem encompassing a Secure Desktop Examination Application, targeted portals for Students, Faculty, and Administrators, an AI-Based Assignment Engine, and a comprehensive Analytics Platform. By standardizing assessments and leveraging natural language processing for personalized remediation, CIRA ensures institutions can reliably track, improve, and predict student placement outcomes.

Problem Statement

Current university placement preparation relies on fragmented assessment tools, manual assignment generation, and disconnected data silos. This prevents real-time tracking of student industry readiness. Faculty lack actionable insights into individual student deficiencies, and students do not receive targeted, adaptive materials to improve their weak areas. The absence of a secure, unified evaluation environment compromises assessment integrity and historical data analysis.

2. Vision, Objectives, & Stakeholders

Vision and Objectives

The vision of CIRA is to democratize and automate placement readiness through data-driven insights and AI-powered personalized learning paths.

- **Objective 1:** Establish a secure, uncompromisable assessment environment for major evaluations.
- **Objective 2:** Automate the identification of student weaknesses and dynamically generate adaptive assignments.
- **Objective 3:** Provide holistic, multi-tiered analytics (Student, Class, Department, Year, Institution) to CIR administrators and faculty.

Stakeholders

Stakeholder	Role in CIRA
University Management	Strategic oversight, institution-level analytics consumer.
CIR Department	Platform administrators, curriculum managers, placement predictors.
Faculty	Quiz conductors, progress evaluators, question bank contributors.
Students	Primary users, assessment takers, assignment consumers.
IT & Security Teams	Deployment, maintenance, audit log monitoring.

3. User Personas & Detailed Workflows

User Personas

Persona	Description	Key Goals
Student (The Learner)	Pre-final/Final year undergraduate.	Access weekly assignments, view progress, take secure exams.
Faculty (The Mentor)	Department instructor guiding students.	Conduct quizzes, adjust assignment difficulty, track class analytics.
CIR Admin (The Strategist)	Placement officer managing readiness.	Analyze institution-wide trends, manage platform configurations.

System Workflow Diagrams (Text Format)

[Student Lifecycle Flow]

1. Semester Start -> Student logs into Secure Desktop Client -> Takes Initial Major Assessment.
2. System calculates baseline Industry Readiness Index (IRI).

3. Student assigned to Readiness Category (Excellent, Average, Poor).
4. Weekly Cycle -> Faculty conducts quiz -> Student completes quiz -> Faculty marks subjective progress -> AI Engine generates personalized PDF assignment.
5. Semester End -> Student takes Final Major Assessment -> System calculates final IRI and placement probability.

4. Requirements & User Stories

Functional Requirements

- The system shall support Excel (.xlsx) bulk imports for question banks.
- The system shall generate personalized weekly assignments downloadable as PDFs.
- The system shall adapt assignment difficulty dynamically based on faculty inputs.
- The system shall categorize students into Excellent, Average, and Poor readiness tiers.

Non-Functional Requirements

- **Performance:** API response times must remain under 200ms for 95% of requests.
- **Scalability:** Support 10,000 concurrent connections during major semester assessments.
- **Availability:** Maintain 99.9% uptime during academic terms.
- **Compliance:** Adhere to educational data privacy standards regarding student academic records.

Detailed User Stories & Use Cases

- **Story 1:** As a Faculty member, I want to override the AI's assignment difficulty for a specific student, so that I can accommodate their unique learning curve.
- **Story 2:** As a Student, I want to download my generated assignment as a PDF, so that I can work on it offline.
- **Story 3:** As a CIR Admin, I want to view a department-wide performance dashboard, so that I can allocate specialized training resources.

5. System Architecture & Technology Stack

Technology Stack Recommendations

- **Frontend Architecture:** React.js for portals, utilizing a unified component library.
- **Backend Architecture:** Node.js (Express) for core microservices, FastAPI (Python) for the AI Generation Engine and Analytics.
- **Database Schema Design:** PostgreSQL for relational student/faculty data; MongoDB (NoSQL) for flexible question banks and JSON assignment structures.
- **Cloud Architecture:** AWS or GCP deployment utilizing managed Kubernetes for orchestration.

Microservice Architecture Components

- User-Service: Authentication and role management.
- Assessment-Service: Exam orchestration, session management, Excel import/export.
- AI-Engine-Service: Natural language processing for assignment generation and evaluation.
- Analytics-Service: Data aggregation and dashboard population.

Deployment Architecture & CI/CD Pipeline

Code committed to the repository triggers automated testing via GitHub Actions. Upon successful unit and integration testing, Docker images are built and pushed to a container registry. Continuous Deployment strategies (Blue/Green deployment) route traffic to the new instances securely without downtime.

6. Secure Examination System Design

Core Specifications

The Secure Desktop Examination Application is designed to ensure zero-tolerance cheating during major assessments.

- **Kiosk Mode & Fullscreen Enforcement:** The application locks the operating system, preventing task switching or minimizing.
- **Browser & Copy-Paste Blocking:** Disables clipboard access, right-click, and external browser launches.
- **Hardware & Device Fingerprinting:** Registers the specific machine MAC address and hardware ID for the session.
- **Multiple Monitor Detection:** Halts the exam if extended displays are connected.

- **AI Tool & Screen Recording Detection:** Continuously scans background processes against a blacklist of recording software and virtual machines.
- **Encrypted Local Storage:** Exam payloads are encrypted at rest on the client and decrypted only in RAM.
- **Violation Logging & Auto-submission:** Minor violations flag the session for proctor review. Severe violations (e.g., unauthorized software detected) trigger immediate auto-submission.

Security Architecture & Threat Model

Network communications utilize strict TLS 1.3 encryption. The threat model accounts for Man-in-the-Middle (MitM) attacks, payload tampering, and credential stuffing. Anti-cheating mechanisms are enforced client-side but validated server-side to prevent bypass via customized binaries.

7. AI Assignment Generation & Readiness Logic

AI Assignment Generation Engine

The FastAPI-driven AI engine utilizes modern NLP models to parse student historical quiz data and major assessment results. It identifies specific granular topics where the student failed. The engine then queries the NoSQL question bank, selecting questions that match the required difficulty and topic, assembling a cohesive assignment structure.

Student Categorization Logic

Students are grouped dynamically based on their evolving scores:

- **Excellent:** Consistently scoring >85% across modules.
- **Average:** Scoring between 60% and 85%.
- **Poor:** Scoring <60%, triggering mandatory faculty review alerts.

Industry Readiness Index Formula

The composite Industry Readiness Index (IRI) is calculated using weighted metrics. Let A_init and A_final be the major assessment scores. Let Q_w be weekly quiz scores for n weeks. Let E_f be faculty subjective evaluations over m weeks.

$$IRI = \left(w_1 \cdot \frac{A_{init} + A_{final}}{2}\right) + \left(w_2 \cdot \frac{\sum_{i=1}^{n} Q_{w,i}}{n}\right) + \left(w_3 \cdot \frac{\sum_{j=1}^{m} E_{f,j}}{m}\right)$$

Weights (w1,w2,w3) are configurable by the CIR administration based on placement season priorities.

8. Data Management & Analytics

Analytics Dashboard Requirements

- **Student Level:** Historical score trendline, topic-wise weakness radar chart.
- **Class Level:** Average IRI, distribution of readiness tiers (Excellent/Average/Poor).
- **Department/Institution Level:** Aggregated metrics for total placement readiness probability.

Excel Import/Export Specifications

- **Import:** Standardized .xlsx templates with strict column validation for Question Type, Content, Options, Correct Answer, Difficulty, and Topic Tags.
- **Export:** Faculty can export weekly results containing Student ID, Name, Score, and auto-calculated percentile.

PDF Generation Requirements

The AI Engine passes structured JSON to a PDF rendering service (e.g., Puppeteer or ReportLab). The generated PDF includes institutional branding, clear question formatting, and unique student identifiers for physical tracking if required.

9. API Specifications & Database Schema

API Specifications (Core Endpoints)

Endpoint	Method	Purpose
/api/v1/auth/login	POST	Authenticates user and returns JWT.
/api/v1/exams/session/start	POST	Initializes secure exam payload.
/api/v1/assignments/generate	POST	Triggers AI engine for a specific student.
/api/v1/faculty/evaluate	PUT	Updates faculty subjective evaluation score.
/api/v1/analytics/department	GET	Returns aggregated JSON data for dashboards.

Database Schema Design (Key Tables)

Table/Collection	Primary Key	Key Attributes
Users (PostgreSQL)	user_id	role, name, email, department, year
Assessments (PostgreSQL)	assessment_id	title, type, start_time, end_time
Results (PostgreSQL)	result_id	user_id, assessment_id, score, metrics_json
QuestionBank (MongoDB)	_id	topic, difficulty, content, correct_option

10. Security, Governance, & Infrastructure

Authentication, Authorization & RBAC

All access is governed by JWT-based authentication. The Role-Based Access Control (RBAC) matrix strictly segregates permissions. Students can only read their data. Faculty can read/write class data. CIR Admins have global read/write access and platform configuration rights.

Monitoring, Alerting, & Audit Logging

Critical actions (grade changes, question bank modifications, policy updates) are stored in an immutable audit log database. Infrastructure monitoring utilizes tools like Prometheus and Grafana, with automated alerts sent to IT staff if API error rates exceed 1% or database latency spikes.

Data Privacy, Backup, & Disaster Recovery

Student assessment data is categorized as highly sensitive. All databases employ AES-256 encryption at rest. Incremental backups occur daily, with full snapshots weekly, stored in a geographically isolated secondary cloud region to ensure Disaster Recovery capabilities.

Risk Assessment

- **Risk:** Faculty adoption resistance to the new platform. **Mitigation:** Comprehensive training sessions and intuitive UI design.
- **Risk:** AI engine generates irrelevant questions. **Mitigation:** Strict metadata tagging in the MongoDB question bank and manual faculty override capabilities.

11. Future Roadmap

Minimum Viable Product (MVP) Scope

The MVP will focus on the core user portals, the deployment of the Secure Desktop Client for major assessments, manual question bank Excel imports, and basic static assignment generation.

Phase-2 Roadmap

- Activation of the NLP-driven AI Assignment Generation Engine.
- Deployment of advanced predictive analytics for placement probabilities.
- Mobile application release for student portal access and notification delivery.